

Homework 3

1. Create a new Vivado project using the files provided for the State Machine Tutorial.
2. Modify the system such that when the consumer receives data, it initiates a new transfer to a second consumer, as shown in Figure 1.

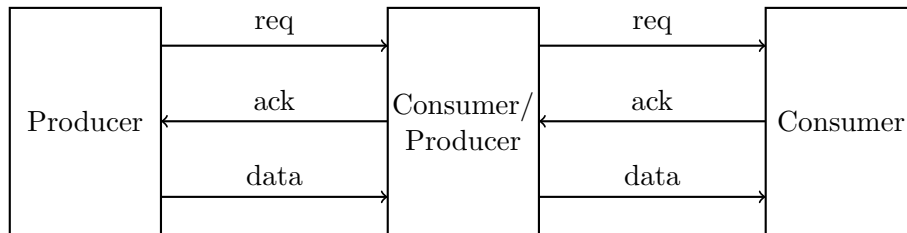


Figure 1: Block diagram of new system

- (a) First, **draw state diagrams** for any new state machines needed to implement the new functionality
 - (b) **Implement the additional component** in a new file, and instantiate the new entity in the `handshakeTop` module such that the `handshakeTop` entity remains the same, and the `o_recv` and `o_data` outputs of the last consumer are the outputs of `handshakeTop`
 - (c) **Make sure that "Consumer/Producer" completes two-way handshake with the Producer and saves received data in an internal register before initiating handshake with the Consumer.**
3. Simulate the system to ensure the data correctly gets sent through all of the blocks. **Include a screenshot of the simulation** with all important signals (including state for all state machines). Ensure that it is clear which modules signals belong to.
 4. Synthesize this design and **record how many flip flops are used and explain in detail where they come all from**
 5. Submit the modified .vhd files and the requested diagrams, simulation screenshot, and synthesis report analysis